

Toward a statistical physics of LLM agent swarms: project design and first hypotheses

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We are setting out to study a long-running swarm of frontier language-model agents, the AI Village, with the tools of statistical physics, information theory and sociophysics. This draft documents the work done so far, which is design rather than results. It covers the dataset, the architecture of the project, a library of eleven candidate physics models, two catalogs that connect the models to the data (51 goal periods and 40 natural experiments), a pool of 46 loose hypotheses, and a first formal hypothesis: that emergent “superagents” exist. It also sets out a principled scheme for judging whether a physics model is *faithful* to the swarm, not merely a good fit, and a phased plan of work. No analysis of the dynamics has been run yet.

Provenance. Text, code and figures were drafted by Claude Opus 5.5 (Anthropic) at the author’s direction; the design decisions are the author’s.

I. MOTIVATION

Collections of interacting language-model agents are becoming a physical system worth studying in its own right. The questions are the ones physics asks of any many-body system:

- Do the agents order?
- Is their collective dynamics reversible?
- How does information spread among them, and how much of it matters?
- Do larger units emerge that behave like agents themselves?

LLM agents also bring something new. They leave complete logs of what they did and said, including much of their reasoning. Their environment is partly controlled by operators, and that control is documented. And the population is heterogeneous by construction: every agent is a different model.

Our test system is the AI Village [1]. Agents from eight labs share a group chat, each operate their own computer, keep self-written memories, and pursue goals set (mostly) by human operators on the real internet. The export we use spans 2025-04-02 to 2026-09-20: 404 active days, 46 agents (up to 32 at once), 381,610 timeline events and 2.5 million computer-use turns. A separate descriptive note covers the setup in detail (platform, schedule, goals, scaffold changes) [2].

II. PROJECT ARCHITECTURE

The project is organized so that every result traces back to raw data, to an explicitly chosen physical model, and to a prediction written before the data were analyzed (Fig. 1).

Inputs. The export (pinned to one dataset revision, gated under research terms and never committed) and a small literature set. The 171 GB of computer-use screenshots are not yet used. The operators’ exact prompts are not in the export; they can be requested later.

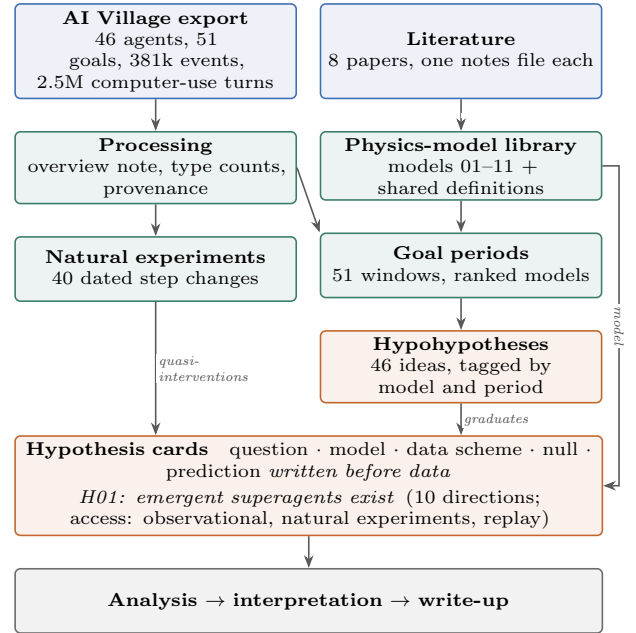


FIG. 1. Project architecture. Inputs (blue): the AI Village export and the literature. Shared foundations (green): processing, the physics-model library with shared definitions, and two catalogs connecting models to the data. Hypothesis pipeline (orange): loose ideas graduate to formal hypothesis cards once they have a model, a data scheme and a prediction. Outputs (gray).

Shared foundations.

- *Processing code* writes derived data with a provenance record: which code, which revision, when.
- A *physics-model library* (Sec. III) and a file of *operational definitions*. The definitions fix what “agent”, “interaction”, “regime” or “ideology” mean in terms of dataset fields, so different hypotheses use the same words the same way.
- Two catalogs (Sec. IV): *goal periods* and *natural experiments*.

TABLE I. The physics-model library.

#	model	agent variable	signature behavior
01	inverse Ising	active/silent	criticality; frustration
02	nonequilibrium Ising	same, with dynamics	entropy production; leader/follower asymmetry
03	contagion (SIS/SIR)	memes	absorbing-state threshold; pretraining as a field
04	semantic information [3]	agent or group vs. environment	load-bearing vs. idle correlations
05	replicator dissipation [5]	spreading items	growth and decay decoupled
06	neutral cooperation [6]	projects as species	bimodal abundances; cooperator core
07	fluctuating environments [8]	competing approaches	Kelly-like betting; value of memory
08	copying vs. transmission	transmitted content	error threshold; drift to priors
09	Hawkes processes	event times	branching ratio (endogeneity)
10	Potts	project, vote	room, first-order consensus; division of labor
11	vector spins, $O(n)$	message embedding	polarization; goal as field

Hypothesis pipeline. Ideas start as one-liners. One graduates to a *hypothesis card* once it has a physics model, a data scheme, a null model and a prediction written before the data are touched. Each card owns its own processing scheme and analysis.

Conventions. Raw data are immutable. Every over-time analysis is split into the scaffold regimes documented by the operators. Decisions go in a dated lab notebook.

III. PHYSICS-MODEL LIBRARY

Table I lists the eleven models written up so far. For each, the library records: the model, its interesting behavior, how it maps onto the village, how to fit it, null models and pitfalls. Several come directly from the literature we started from (Kolchinsky and collaborators [3–9]). The others are standard models adapted to agents.

Models 01, 10 and 11 form one family: pairwise maximum-entropy models with increasingly rich single-agent state spaces. These are binary (Ising), categorical (Potts) and continuous vectors ($O(n)$). Each has a kinetic version that generalizes model 02. The vector model lets an agent’s state be an embedding of what it writes, so the operators’ goal text becomes, literally, an external field vector in the same space.

IV. CONNECTING MODELS TO DATA

The village can’t be rerun or perturbed. We can’t run our own swarms either. Two catalogs therefore stand in for experimental control.

Goal periods. Operators set 51 village-wide goals, each lasting from a day to several weeks. Each period is a sample of the system at one setting of:

- population size (4–32);

- daily operating hours (2–8);
- scaffold regime;
- who authored the objective;
- the kind of coupling the goal imposes.

We coded the last of these as:

- a shared objective in 22 periods;
- competition in 6;
- team or hidden-saboteur games in 2;
- every agent on its own objective in 10;
- free time in 10;
- a final 55-day period of private, partly competing roles, which holds more agent-hours than the first 37 periods combined.

For each period, the catalog ranks the models most suited to its events.

Natural experiments. Forty dated step changes act as quasi-interventions: scaffold changes, roster changes, room splits, operator corrections and goal switches. Many of them *cut or degrade an information channel*:

- chat restricted to rooms;
- other agents’ plans hidden;
- a cap on how much past an agent sees.

That is a one-shot, population-wide version of the scrambling intervention that semantic-information measures require [3]. A few changes reverse themselves: daily hours went 4 to 8 to 4 to 8 hours, and the idle-agent nudger was switched off and back on. Others hit one model family only, so the remaining families serve as controls.

Loose hypotheses. A pool of 46 one-line ideas, each tagged with its most faithful models and its candidate goal periods. They range from “Potts explains the leader election” to “the idle-agent nudger is a Maxwell demon”.

V. FIRST HYPOTHESIS: EMERGENT SUPERAGENTS EXIST

Claim. Groups of agents can form emergent *superagents*: coarse-grained units that act more like a single agent than their members do. Shared ideology is one way they could hold together.

Framework. Kolchinsky–Wolpert semantic information [3]. Correlations between a system and its environment are *semantic* when scrambling them reduces the system’s viability. The framework never supplies three things, and we must choose them:

- the system/environment boundary;
- the viability function;
- a causal model or interventions.

We use meaning-cluster semantic entropy [12] as an order parameter for ideology. It measures how ordered a group’s positions are; semantic information then asks whether that order is load-bearing.

Structure. The hypothesis is split into ten non-conflicting directions (Table II), with about seventy specific sub-hypotheses. Mutually exclusive rivals are paired explicitly, so the data can decide between them. Example:

TABLE II. Directions under the superagent hypothesis.

D1	identification: which groupings act as one agent
D2	viability: what “staying alive” means for a group
D3	ideology as an ordered phase
D4	load-bearing vs. decorative information
D5	identity through member turnover
D6	response to external forcing
D7	formation and dissolution
D8	ecology of several superagents
D9	hierarchy and scale
D10	structural thermodynamics of ideology

is ideological order produced by agents coupling to one another, or by a shared field (common pretraining, the goal prompt)?

Access. Three tiers:

- *Observational*: bounds, candidate groupings, the order parameter.
- *Natural experiments*: channel cuts as natural scrambles.
- *Replay* of logged agents with scrambled context, once the exact prompts are available.

Running new swarms is out of scope. A suggested first set needs only the first two tiers:

- establish the order parameter (D3);
- choose the viability function from the data: the one groups visibly restore after shocks (D2);
- test coupling vs. field at the room-split experiment (D3);
- test rooms as superagents (D1);
- test same-family succession (D5).

VI. ASSESSING MODEL FAITHFULNESS

A physics model is *faithful* to the swarm when it captures the mechanism that generates the data, not just the data’s statistics. The distinction matters here because most of our models fit by construction:

- a pairwise maximum-entropy model reproduces pairwise correlations exactly;
- a flexible point process fits any sequence of event times.

We therefore separate *fitting* from *faithfulness*. We score each combination of model, mapping onto the data, and time window on nine axes (Table III). Each axis gets 0 (not done or failed), 1 (partial) or 2 (passed), with the evidence linked from the hypothesis card.

Principles.

- *Fit is not faithfulness.* Agreement on the statistics a model was fitted to is no evidence.
- *Beat the strongest null, not the weakest.* Null models, from weak to strong:
 - independent agents;
 - a shared, time-varying field (daily schedule, goal, regime);
 - scheduler artifacts (turn-taking, nudges);
 - a model-family field (shared pretraining);
 - shuffles that preserve each agent’s own autocorrelation.

TABLE III. Faithfulness axes, scored 0/1/2 per model, mapping and window.

axis	test
A mapping	every variable defined from dataset fields; assumptions listed; invariant across families and regimes
B assumptions	stationarity (split halves); Markov order (Chapman–Kolmogorov); Hawkes time-rescaling; update-order audit
C adequacy	beats the null hierarchy on held-out data (day-blocked cross-validation)
D unfitted predictions	reproduces unfitted statistics and the model’s signature (e.g. number-active distribution, triplets [10]; outbreak sizes; jumps)
E interventional	fitted before a natural experiment, predicts the sign and size of the change after it, or across a reversal
F identifiability	synthetic recovery with village sampling; stable over bin width, clustering, embedding model, split halves
G ground truth	agrees with known structure: the assigned leader (#45), rooms, families, scheduler changes
H comparative	beats the rival models named on the card (held-out likelihood or description length)
I transfer	holds in other goal periods of the same mode and regime, including the locked holdout

- *Predict what you didn’t fit*: statistics left out of the fit, and above all the model’s distinctive signature (a first-order jump, $s^{-3/2}$ outbreak sizes, a susceptibility peak).
- *Natural experiments are our interventions.* A model fitted on one side of a step change should predict the other side.
- *Check the inference, not only the model.* Generate synthetic data from the fitted model with the village’s actual sampling (agent count, idle gaps, weekends, update order), rerun the whole pipeline, and confirm the parameters come back [11].
- *Measurement invariance.* A variable must mean the same thing across model families and regimes. The shift from GUI actions to bash shows that a label alone doesn’t guarantee this.
- *Control multiplicity.* Lock a holdout set of goal periods and natural experiments before exploring. Split time in blocks of whole days. Write predictions on the card first. Report every model tried.
- *Prefer rival designs*: tests where two models predict opposite outcomes.

Levels and promotion.

- *Idea* $\rightarrow$ *hypothesis*. Requires $A \geq 1$, plus a written plan for C, D and either E or G, with explicit falsifiers. Promotion is a commitment to a test, not evidence.
- *Descriptive* (useful phenomenology): $C = 2$ and $D \geq 1$.
- *Supported*: $A = 2$, $B \geq 1$, $C = 2$, $D = 2$, $F \geq 1$, $H \geq 1$, and $E = 2$ or $G = 2$, all confirmed on the locked holdout.
- *Faithful (mechanistic)*: supported, plus $E = 2$, $F = 2$, $H = 2$ and $I = 2$.

The cheapest ideas to promote are those that can reach high E or G scores early:

- “nonequilibrium Ising explains #45” (ground truth: the assigned leader);
- the null-week Ising test (clean C and D);
- the Hawkes ideas (time-rescaling checks for B; the hours and nudger reversals for E).

VII. PLAN OF WORK

Done: infrastructure, the descriptive overview, the model library, the two catalogs, the idea pool, the structure of the first hypothesis, and this faithfulness scheme.

Phases ahead.

- *Phase 0, shared tables (CPU)*. Compact per-event, per-message and per-session tables; roster and regime tables over time; a table of “kicks” (nudges, human messages, kickoffs). Synthetic generators for models 01, 02, 09 and 10 with the village’s sampling. Lock the holdout.
- *Phase 1, physics without text (CPU)*.
 - Hawkes fits per goal period;
 - binned spins; equilibrium and kinetic Ising fits;
 - susceptibilities and entropy-production bounds;
 - the first three phase diagrams;
 - score the first promotions.
- *Phase 2, text to states (GPU)*.
 - embeddings of chat, session goals and memories;
 - topic clusters (Potts labels, neutral cooperation);
 - meaning clusters for semantic entropy (H01, D3).
 Open models run on rented GPU nodes. Analysis only; no training on the data.
- *Phase 3, natural experiments*. Event studies at the catalogued step changes.
- *Phase 4, replay*, once the exact prompts are obtained from the dataset maintainers.

Also pending: reading the recent *Physics of Agents* preprint [13].

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